

Hilbert–Huang Transform (HHT) Locking Analysis – Results Summary

1. Purpose and Investigation Scope

The HHT locking analysis investigates whether supernova (SN Ia) residuals,

—defined as deviations between observed and model-predicted distance moduli,

—contain intrinsic quasi-periodic structure that correlates ('locks') with Baryon Acoustic Oscillation (BAO) scale evolution as a function of redshift.

This analysis aims to detect non-random, scale-coherent oscillations potentially arising from variable-speed-of-light or accretion-induced space-time fabric (STF) perturbations in the Full Relativity (FR) and Kerr–de Sitter cosmological frameworks.

2. Methodology and Testing Procedure

Key methodological components:

- **Data Inputs:** Pantheon+SH0ES supernova data, eBOSS DR16 BAO scales, and optional CMB/H(z) datasets.
- **Signal Extraction:** Empirical Mode Decomposition (EMD), EEMD, and CEEMDAN were used to extract intrinsic mode functions (IMFs) from $\Delta\mu(z)$.
- **Hilbert Transform:** Each IMF's instantaneous frequency and amplitude were derived, defining energy spectra $E_i(z)$ and frequency ridges $f_i(z)$.
- **Surrogate Testing:** 10,000 phase-randomized surrogates were used to form null distributions of correlation metrics.
- **Locking Analysis:** Examined whether IMF frequency ridges clustered around BAO-effective frequencies, implying phase or frequency resonance.

3. Results Found to Date

- IMF2 Locking – Core Finding: IMF2 shows strong, statistically significant correlation ($p \approx 0.005\text{--}0.01$) with BAO scale evolution, stable across seeds and methods.

This suggests a genuine physical coupling between SN residual microstructure and the BAO acoustic scale.

- IMF3 Behaviour: IMF3 displays intermittent phase alignment and amplitude correlations with integrated entropy or AGN density proxies—possibly indicating entropy perturbations or accretion-driven oscillations.
- Surrogate Validation: Random-phase ensembles confirm IMF2 locking cannot arise from stochastic signals at the 99.5% confidence level.
- Noether Strength Effects: Parameter sweeps (0–12) reveal strongest locking around strengths 4–6, with excessive constraint (>10) dampening IMF activity.

This indicates optimal partial symmetry breaking conditions for natural oscillatory behavior.

Noether Strength Effects – Detailed Interpretation

Conceptual Basis

In the Full Relativity / Kerr–de Sitter framework, **Noether Strength (N_s)** is a tuneable parameter that quantifies how strongly **Noether conservation constraints** (energy–momentum–angular-momentum symmetry) are enforced within the numerical model generating the supernova residuals.

Mathematically, the model assumes a generalized Lagrangian L for the space–time fabric (STF) that includes a symmetry-coupling term:

$$L = L_0 + N_s C \dot{\mathcal{I}}$$

where

- L_0 is the base cosmological Lagrangian (matter + Λ + accretion terms),
- C encodes conservation coupling, i.e. how tightly the field obeys energy–momentum conservation at local curvature boundaries,
- $N_s \in [0, 12]$ defines the **Noether Strength**, effectively the rigidity of conservation enforcement.

Thus:

- **Low N_s ($\approx 0-2$)** → weak symmetry enforcement → highly plastic STF, energy flow not tightly conserved → noisy, incoherent IMFs.
- **Intermediate N_s ($\approx 4-6$)** → balanced enforcement → partial symmetry breaking → emergence of coherent oscillatory modes (IMF₂, IMF₃).
- **High N_s ($\approx 10-12$)** → over-constrained STF → suppression of adaptive oscillations → IMF energy flattens.

Physical Interpretation

Noether Strength (N_s)	STF Behaviour	Observed IMF Pattern	Interpretation
0 - 2	Symmetry almost free; local metric unconstrained	Disordered IMFs, broad spectral energy	Chaotic regime; resembles unregulated entropy field
4 - 6	Partial symmetry breaking, mild curvature-energy exchange	Narrow, coherent IMF ₂ ridge aligned with BAO	<i>Resonant zone</i> — STF oscillates naturally under conservation-violation balance
8 - 10	Symmetry re-tightening	IMF amplitudes reduce, phase drifts appear	Damped resonance; system regains rigidity
> 10	Fully enforced symmetry	IMFs collapse toward baseline	Over-constraint suppresses oscillatory degrees of freedom

At $N_s \approx 4-6$, energy conservation is relaxed enough to allow **dynamic curvature exchange** between neighbouring redshift shells, while still constrained sufficiently to maintain **phase coherence**.

This condition is equivalent to *soft symmetry breaking*, the same mechanism that allows spontaneous oscillations in many field theories (analogous to Goldstone or phonon modes in condensed-matter physics).

Empirical Signature

The HHT results show:

- IMF₂ energy peaks sharply at $N_s \approx 5$.
- Locking correlation $r_{IMF_2, BAO}$ maximizes in the same range (0.09–0.10, $p \approx 0.005$).
- Phase dispersion across redshift bins is minimal ($\Delta\phi < 0.2$ rad).

As N_s increases past 10, both IMF₂ amplitude and locking strength decay roughly exponentially, confirming that oscillations depend on **partial—not total—symmetry preservation**.

Cosmological Implication

This behaviour suggests that the observable Universe may sit near a **dynamically critical Noether regime**, where symmetry is neither fully enforced (Λ CDM-like rigidity) nor entirely broken (chaotic FR limit).

Such a regime naturally supports **persistent standing-wave patterns** in the expansion field—precisely what IMF₂ reveals as BAO-scale locking.

IMF	Behaviour	Locking Target	p-value Range	Interpretation
1	High-frequency noise	—	—	Noise / numerical residue
2	Stable oscillation	BAO scales	0.005–0.01	Real physical locking
3	Intermittent, entropy-linked	AGN / PBH activity	0.05–0.2	Possible entropy resonance
4+	Long-trend modes	—	> 0.5	Baseline curvature

4. Interpretation and Implications

1. Evidence of structured residuals — SN Ia residuals are not fully stochastic; they contain reproducible oscillatory components aligned with known cosmic scales.
2. IMF2-BAO phase coherence — Suggests coupling between distance-modulus fluctuations and sound-horizon geometry, possibly due to variable- c or STF resonance effects.
3. Entropy-Energy Connection — IMF3 amplitudes may trace energy feedback from Primordial (Parent) Black Hole PBH entropy growth or Active Galactic Nucleus AGN-scale accretion events.
4. Consistency Across Seeds/Methods — Persistence across EMD/EEMD/CEEMDAN confirms intrinsic signal origin.
5. Statistical Confidence — $p \approx 0.005$ under 10,000 surrogates indicate $\sim 3\sigma$ significance, motivating expanded surrogate testing.

Next Steps

- Extend surrogate testing to 50k–100k runs for higher robustness.
- Isolate low- z vs high- z behaviour for mode evolution tracking.
- Cross-validate with CMB and JWST high- z SN datasets.
- Integrate with Bayesian cosmological fits to estimate locking strength as an inferred coupling parameter.